

- 7 **(a)** **(i)** Find the modulus and argument of z_1 , where $z_1 = 1 + i$. **[2]**
- (ii)** Given that $|z_2| = 2$ and $\arg(z_2) = \frac{1}{6}\pi$, express z_2 in $a + bi$ form, where a and b are exact real numbers. **[2]**
- (b)** Using these results, find the exact value of $\sin \frac{5}{12}\pi$, giving the answer in the form $\frac{\sqrt{m} + \sqrt{n}}{p}$, where m , n and p are integers. **[5]**